Marta Grobelna

Doctoral Researcher, Algorithm Development and Optimization

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Profile

Doctoral researcher at the Technical University of Munich developing algorithms for concurrent stochastic games and Markov decision processes with formal precision guarantees. I independently developed, proved correct, and implemented a novel value-iteration algorithm in the Java-based PRISM model checker, using linear and quadratic programming (AMPL, Gurobi) as core tools. At Fraunhofer IKS, I translated industry customers' goals (BMW, Huawei) into mathematical models and working prototypes, and I maintain and migrate a production Python web platform (Automata Tutor). Next, I want to bring this experience to industry and tackle challenging real-world problems.

Experience

Technical University of Munich (TUM), Doctoral Researcher, Algorithm Development — Munich, Germany — Sept 2022 – present — 4 years 2 months

- Design algorithms for stochastic games and Markov decision processes (MDPs), combining mathematical analysis, correctness proofs, software implementation, and computational evaluation.
- Independently designed, proved correct, and implemented in Java the first value-iteration algorithm for concurrent stochastic games with guaranteed precision for maximum-reachability objectives, and integrated it into the PRISM probabilistic model checker (LICS 2025).
- Formulated optimization problems arising from concurrent stochastic games in AMPL and solved them using Gurobi and other optimizers through the AMPL Java API.
- Co-authored a paper on hot-starting and incremental value iteration for MDPs that reuses previous results when a model changes, with formal precision guarantees (under review).
- Currently developing bounded strategy iteration and a new quadratic-programming formulation for concurrent stochastic games, and extending the methods to partially observable MDPs (POMDPs).

Fraunhofer Institute for Cognitive Systems (IKS), Research Associate, Applied R&D for Autonomous Systems — Munich, Germany — Dec 2019 – Aug 2022 — 2 years 9 months

- BMW project: worked directly with the customer to turn the question "is this overtaking maneuver safe?" into a precise mathematical problem (reachable-set computation from vehicle dynamics via flow-pipe construction, checking safety constraints specified in temporal logic for hybrid automata). Delivered a software prototype and was first author of the resulting EDCC 2021 paper.
- Huawei project: analyzed safety properties of sensor-fusion algorithms for autonomous driving and contributed to the white paper *Trustworthy AI for Intelligent Traffic Systems*.
- Coordinated work packages, clarified customer requirements, and presented results to industry stakeholders.

RWTH Aachen University, Student Assistant, Software and Verification Tools — Aachen, Germany — Oct 2018 – Aug 2019 — 11 months

- Benchmarked verification tools and contributed C++ implementation and project documentation.

Technical Skills

Programming: Python (primary), Java, C++, MATLAB

Optimization: Linear programming (LP), quadratic programming (QP), AMPL, Gurobi and other solvers, hot-starting and incremental algorithms

Modeling: Stochastic games, Markov decision processes (MDPs), value and strategy iteration

Engineering: Git, CI/CD, pytest, Flask, MySQL, Docker, Linux, Bash, integration into existing codebases, automated testing, AI-assisted development (Claude Code)

Domain expertise: Formal verification, probabilistic model checking, multi-agent systems, safety-critical autonomous systems, partially observable MDPs (POMDPs)

Projects

Automata Tutor 2022 – present

Web-based platform for automated assessment of exercises in automata theory (Python/Flask, MySQL, pytest, CI/CD).

- Re-implemented the entire frontend in Python/Flask and migrated the database from H2 to MySQL, including a redesign of the SQL schema.
- Built the automated test infrastructure (pytest fixtures, mock databases) within a collaborative Git and CI/CD workflow.
- Currently guiding students through the migration of the backend.

Publications

Stopping Criteria for Value Iteration on Concurrent Stochastic Reachability and Safety Games 2025

Marta Grobelna, Jan Křetínský, Maximilian Weininger

LICS 2025

Dynamic Risk Management for Safely Automating Connected Driving Maneuvers 2021

Marta Grobelna, Joao-Vitor Zacchi, Philipp Schleiss, Simon Burton

EDCC 2021

Trustworthy AI for Intelligent Traffic Systems (ITS) 2021

Stefano Bortoli, Margherita Grossi, Simon Burton, **Marta Grobelna**, Yuki Hagawara, Philipp Schleiß

Fraunhofer IKS white paper

Education

PhD **Technical University of Munich**, Computer Science — Munich, Germany

- Expected completion in February 2027. — Sept 2022 – present

M.Sc. **RWTH Aachen University**, Computer Science — Aachen, Germany

- Master's thesis: Integration of hybrid systems verification methods in Matlab — 2017 – 2019

B.Sc. **RWTH Aachen University**, Computer Science — Aachen, Germany

- Bachelor's thesis: Solving Pseudo-Boolean Constraints — 2013 – 2017

Leadership and Teaching

- Coordinated approximately 15 teaching assistants for a university course with 750+ students.
- Supervised four Bachelor's and two Master's theses.
- Regularly present my research at international conferences.

Languages

German: Native

Polish: Native

English: Fluent

Spanish: Basic

Norwegian: Basic

Personal Interests

Bouldering: Regular sessions at my home climbing gym.

Cycling: Long rides in nature and a multi-day bike trip with friends or family almost every year.

Hiking: Long, demanding mountain hikes and multi-day trekking trips.

Music: Playing guitar since I was 13 and drums for two years, mostly metal.

Kitesurfing: One of my favorite water sports.

Reading: Fantasy novels, books about expeditions, and non-fiction on history and science.